Project Proposal

Pharmacy Management System – Desktop Application

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1. Introduction

This proposal presents a milestone-based plan to design, develop, install and hand over a desktop Pharmacy Management System for a single pharmacy run by a doctor. The client referenced the D Watson pharmacy system; the scope below covers the parts of that kind of system that a single pharmacy uses every day. Two options are quoted separately:

- **Option 1 – Purchasing & Stock Management:** suppliers, purchases, batch/expiry stock, stock removal and expiry/low-stock alarms.
- **Option 2 – Purchasing + Sales & Billing:** everything in Option 1, plus a counter sales (POS) screen, printed bills and sales history.

Every optional add-on is priced individually in Section 6, so the client can start small and add features later.

2. Project Scope

2.1 Option 1 – Purchasing & Stock Management

Note: Anything not listed under "Included" is not part of Option 1.

Included Modules – Option 1:

1. Setup & Access

Flow 1: First-run setup (pharmacy name, address, licence no., logo for printouts)
Flow 2: Owner login with password, and password change
Flow 3: Basic manual backup to a folder / USB drive and restore

2. Medicine Catalogue

Flow 1: Add / edit / deactivate a medicine (brand name, generic/salt formula, strength, form, company, category)
Flow 2: Search medicines by brand name or by generic/salt (shows substitutes with the same formula)

Flow 3: Set a minimum stock level and an expiry warning period per medicine (or use the global default)

3. Company & Supplier Records

Flow 1: Add / edit medicine companies (manufacturers)
Flow 2: Add / edit suppliers / distributors (contact person, phone, address, NTN if any)
Flow 3: View the purchase history of a supplier

4. Purchases & Stock-In

Flow 1: Record a purchase: supplier, supplier invoice no., invoice date, received date
Flow 2: Add purchase lines: medicine, batch no., expiry date, quantity, bonus quantity, purchase price, MRP / sale price
Flow 3: Save the purchase, which adds the stock to the matching batches automatically
Flow 4: View, search and reprint (A4) a recorded purchase; edit it with the change logged

5. Stock View & Search

Flow 1: Current stock balance per medicine (total of all batches)
Flow 2: Batch details: batch no., expiry, quantity left, purchase price, supplier
Flow 3: Filter stock by company, category, supplier, expiry range or low stock

6. Stock-Out & Adjustments (how stock is reduced without a sales module)

Flow 1: Stock Issue: record medicine used / dispensed (e.g. in the clinic), batch picked earliest-expiry-first
Flow 2: Write-off: record damaged, expired or missing stock with a reason
Flow 3: Physical stock count: enter the counted quantity, and the system posts the difference as an adjustment
Flow 4: Stock movement history (ledger) per medicine and batch, showing every in, out and adjustment

7. Expiry & Low-Stock Alarms

Flow 1: Dashboard tiles: Expired, Expiring Soon, Low Stock, Out of Stock
Flow 2: Flashing red/amber highlights on affected medicines across stock screens
Flow 3: Sound alarm on app start and once per day while alerts are open (can be muted or snoozed). **Sound alarm is included.**
Flow 4: Settings: global expiry warning period (default 6 months, to be confirmed with the client) and default minimum stock level

Excluded Modules – Option 1:

1. Sales & Billing => "Included in Option 2"

2. Every item in Section 6 (Optional Add-ons) => "Priced separately"

2.2 Option 2 – Purchasing + Sales & Billing

Note: Includes every module in Option 1 plus the modules below.

Included Modules – Option 2 (in addition to Option 1):

8. Counter Sale (POS)

Flow 1: Search and add medicines to the bill by brand name or by generic/salt
Flow 2: Enter quantity; the sale price is pre-filled and can be edited by the owner
Flow 3: Automatic bill total, cash received and change returned
Flow 4: Expired batches are blocked from sale; a warning shows when stock is insufficient

9. Batch Deduction

Flow 1: Sold quantity is deducted from the earliest-expiring batch first (FEFO), splitting across batches when needed
Flow 2: Stock alarms update immediately after each sale

10. Bills & Receipts

Flow 1: Print a bill on an 80 mm thermal receipt printer or on A4
Flow 2: Save every bill with a sequential bill number
Flow 3: Search and reprint any past bill
Flow 4: Cancel (void) a bill by the owner, which restores the stock and logs the reason

11. Sales History & Daily Totals

Flow 1: Sales history with filters (date range, bill no., medicine)
Flow 2: Daily sales total and number of bills; day-end summary print

Excluded Modules – Option 2:

1. Discounts, taxes, customer credit, returns, staff logins => "Optional add-ons (Section 6)"

2. FBR POS / e-invoicing integration => "Optional add-on (Section 6)"

3. Technology Stack & Platform

- **Application:** Flutter Desktop, a native Windows app (.exe installer)
- **Database:** SQLite (local, encrypted at rest), stored on the pharmacy computer
- **Supported OS:** Windows 10 and Windows 11 (64-bit). macOS can be built on request.
- **Offline:** Yes. The system works fully offline; internet is needed only for the remote/cloud add-ons.
- **Printing:** 80 mm thermal receipt printers (ESC/POS) and any Windows A4 printer
- **Minimum PC:** Intel Core i3 / 4 GB RAM / 10 GB free disk, 1366×768 screen
- **Language:** English interface (Urdu printouts can be quoted on request)

4. Project Timeline

4.1 Option 1 – 6 Weeks

Modules	Wks	Design Screens	Desktop Dev	Database / Logic
Foundation & Setup	1	• App shell & design system • Setup wizard & login • Dashboard layout	• Windows app skeleton • Navigation & theming • Installer project	• SQLite schema & migrations • Encryption • Backup / restore
Catalogue & Suppliers	2	• Medicine form & list • Company / supplier screens	• CRUD screens • Salt/generic search	• Medicine, company, supplier tables • Search index
Purchases & Stock-In	3-4	• Purchase entry grid • Purchase list & detail • Stock & batch views	• Keyboard-first purchase entry • A4 purchase print	• Batch / expiry model • Stock ledger • Edit logging
Stock-Out & Alarms	5	• Stock issue / write-off • Physical count • Alert tiles & highlights	• Flashing indicators • Sound alarm, snooze & mute	• FEFO batch picking • Alert rules & thresholds
UAT & Handover	6	• Design QA	• Bug fixes • Release build	• Testing with the client's real data • Installation & training

4.2 Option 2 – 9 Weeks

Weeks 1–5 are the same as Option 1, then:

Modules	Wks	Design Screens	Desktop Dev	Database / Logic
Counter Sale (POS)	6-7	• POS screen • Bill preview	• Fast keyboard / mouse billing • Insufficient-stock & expired-batch guards	• Sales model • FEFO deduction across batches
Bills & Sales History	8	• Thermal & A4 bill layouts • Sales history & day-end	• Printing & reprint • Void bill	• Bill numbering • Daily totals
UAT & Handover	9	• Design QA	• Bug fixes • Release build	• Testing with real data • Installation, printer setup & training

5. Project Payment Schedule

All prices are in **USD**, are one-time, and exclude applicable taxes (if any). There is **no recurring licence fee** for the core system. PKR payment can be agreed at the exchange rate on the invoice date.

5.1 Option 1 – Total: USD 2,000 | Delivery: 6 weeks

Milestone	Duration	Price Quote	Modules Covered	Acceptance
Advance (15%)	On signing	USD 300	Requirement sign-off, data sample, setup	Signed scope
Milestone 1	Weeks 1–2	USD 500	Foundation & Setup, Catalogue & Suppliers	Demo build approved
Milestone 2	Weeks 3–4	USD 600	Purchases & Stock-In, Stock View	Demo build approved

Milestone	Duration	Price Quote	Modules Covered	Acceptance
Milestone 3	Weeks 5–6	USD 600	Stock-Out & Alarms, UAT, installation, training	Installed at pharmacy
Total	**6 weeks**	**USD 2,000**		

Calculation: $300 (advance) + $500 (M1) + $600 (M2) + $600 (M3) = **USD 2,000**

5.2 Option 2 – Total: USD 3,200 | Delivery: 9 weeks

Extra cost to add Sales & Billing to Option 1: USD 1,200 (+3 weeks)

Milestone	Duration	Price Quote	Modules Covered	Acceptance
Advance (15%)	On signing	USD 480	Requirement sign-off, data sample, setup	Signed scope
Milestone 1	Weeks 1–2	USD 500	Foundation & Setup, Catalogue & Suppliers	Demo build approved
Milestone 2	Weeks 3–4	USD 600	Purchases & Stock-In, Stock View	Demo build approved
Milestone 3	Week 5	USD 450	Stock-Out & Alarms	Demo build approved
Milestone 4	Weeks 6–7	USD 700	Counter Sale (POS), batch deduction	Demo build approved
Milestone 5	Weeks 8–9	USD 470	Bills & Sales History, UAT, installation, printer setup, training	Installed at pharmacy
Total	**9 weeks**	**USD 3,200**		

Calculation: $480 (advance) + $500 (M1) + $600 (M2) + $450 (M3) + $700 (M4) + $470 (M5) = **USD 3,200**

Advance Payment

- **Advance Payment:** 15% of the selected option is paid upfront.
- **Purpose:** Covers requirement sign-off, database design and project setup.
- **Cancellation:** If the client decides not to proceed, the advance is retained for the preparatory work done.
- **Milestone payments:** Each milestone is invoiced when its demo build is approved. The next milestone starts after payment.
- **Change requests:** Features outside this scope are quoted separately before work starts.

6. Optional Add-ons (Priced Separately)

These are optional and not confirmed. Each add-on is a one-time price added to the milestone in which it is delivered. "Needs Option 2" means the add-on depends on the Sales & Billing module.

Add-on	What it includes	Extra cost	Time	Needs Option 2?
Supplier accounts	Payments to suppliers, outstanding balances, supplier statement (print / PDF)	USD 350	+1 wk	No
Returns & stock corrections	Supplier returns linked to purchase/batch; customer returns against a bill. (Damaged/missing write-offs are **included** in Option 1.)	USD 300	+1 wk	Customer returns only

Add-on	What it includes	Extra cost	Time	Needs Option 2?
Barcode scanning	USB/Bluetooth scanner for stock entry and sales; map one or more barcodes per medicine	USD 200	+0.5 wk	Sales scanning only
Barcode label printing	Design and print shelf/medicine labels (name, price, batch, expiry, barcode)	USD 250	+0.5 wk	No (needs label printer)
Pack & unit selling	Box → strip → tablet conversion; buy in boxes, sell in strips or tablets with correct stock	USD 400	+1 wk	Unit selling only
User accounts	Separate staff logins, role permissions (Owner / Salesperson / Store), audit log of changes	USD 350	+1 wk	No
Backup & restore (automatic)	Scheduled automatic backups to a second drive/USB, retention, one-click restore	USD 200	+0.5 wk	No
Reports & exports	Stock value, purchases, near-expiry, dead stock, sales and profit reports; Excel & PDF export	USD 450	+1 wk	Sales & profit reports only
Import existing records	Excel template, validation and import of medicines, opening stock and batches	USD 200	+0.5 wk	No
Customer accounts	Customer records, credit sales, payments received, outstanding balances	USD 350	+1 wk	Yes
Discounts & taxes	Item and bill discounts, configurable tax rates on bills	USD 250	+0.5 wk	Yes
Prescription printing	Doctor's prescription entry with patient name/age, medicine, dose, duration; print on letterhead	USD 300	+1 wk	No (send-to-bill needs Option 2)
Multiple computers (LAN)	One computer acts as the server, others connect over the local network with shared records	USD 700	+1.5 wks	No
Multiple branches	Separate stock per branch, branch transfers, consolidated view	USD 1,200	+3 wks	No (needs Remote/cloud)
Remote / cloud access	Cloud database and sync; owner can view stock and sales remotely	USD 1,400	+3 wks	No

6.1 Other Suggested Add-ons

Add-on	What it includes	Extra cost	Time	Needs Option 2?
Controlled drugs register	Separate register for narcotic / psychotropic medicines with prescription reference, as required by drug inspectors	USD 250	+0.5 wk	No
Reorder suggestions	Suggested purchase list from low stock and recent usage; print / send to supplier	USD 250	+0.5 wk	No
Owner dashboard	Charts for sales, stock value, expiring value and top medicines	USD 200	+0.5 wk	Sales charts only
FBR POS integration	Report each bill to FBR and print the FBR invoice number / QR on the bill (if the pharmacy is required to integrate)	USD 600	+1.5 wks	Yes (and Discounts & taxes)

Note: Multiple computers, Multiple branches, Remote/cloud access and Pack & unit selling change the database design. They should be decided before Milestone 1. Adding them after delivery may cost more than the prices above.

7. Deliverables

- Windows installer (.exe) of the Pharmacy Management System for the selected option and add-ons
- Installation on the pharmacy computer (on-site in the same city or remote via AnyDesk)
- Printer setup: receipt printer and A4 printer (and label printer / scanner if those add-ons are taken)
- Training: one session (up to 2 hours) for the owner and staff, plus a short user guide (PDF)
- Complete source code and database schema handed over after the final payment

8. Ownership, Warranty & Support

- **Source code ownership:** The client owns the source code and data after the final payment.
- **Bug fixes:** Free bug fixes for **60 days** after go-live for anything that does not work as described in this proposal.
- **Support after warranty (optional):** USD 30 / month or USD 300 / year for bug fixes, remote help and minor changes (up to 4 hours / month).
- **New features after delivery:** Quoted separately.

9. Client's Support

To ensure timely delivery, the following is expected from the client:

Information & Data:

- Pharmacy name, address, licence no. and logo for printouts
- A sample of the current medicine list, suppliers and a few purchase invoices
- Confirmation of the expiry warning period (e.g. 6 months) and default minimum stock level
- One contact person who can test and approve each milestone within 3 working days

Hardware & Licences:

- A Windows 10/11 computer with a valid Windows licence
- Printers and scanners as required (see Appendix A); purchased by the client
- For Remote/cloud add-ons: an internet connection and a cloud hosting account

Appendix A – One-time & Recurring Costs (Third-Party)

These are not part of the development price. Prices are indicative and depend on brand and market.

Item	Needed for	Type	Indicative cost
80 mm thermal receipt printer	Option 2 bills	One-time	PKR 12,000 – 25,000
USB barcode scanner	Barcode scanning add-on	One-time	PKR 4,000 – 12,000
Barcode label printer + label rolls	Label printing add-on	One-time + consumables	PKR 25,000 – 50,000
USB drive / external disk	Backups	One-time	PKR 3,000 – 10,000
UPS for pharmacy PC	Recommended (protects data during load-shedding)	One-time	PKR 15,000 – 40,000
Cloud server / database	Remote/cloud & Multiple branches add-ons	Recurring	USD 10 – 25 / month
Core system licence	Option 1 / Option 2	—	None (no recurring fee)

Appendix B – Answers to Quotation Questions

Question	Answer
Currency	USD (PKR at the invoice-date rate on request)
One-time vs recurring cost	Development is one-time. No recurring cost unless Remote/cloud add-ons or paid support are taken.
Supported desktop OS	Windows 10 / 11 (64-bit)
Works offline?	Yes, fully offline
Sound alarm included?	Yes, with flashing highlights, mute and snooze
Stock reduction without sales (Option 1)	Stock Issue, Write-off and Physical Count screens (Section 2.1, Module 6)
Installation / printer setup / training	Included (Section 7)
Source code ownership	Client, after final payment
Bug fixes / support	60 days free; optional support plan afterwards
Payment terms	15% advance, then per milestone (Section 5)
Delivery time	Option 1: 6 weeks. Option 2: 9 weeks. Add-ons add the time shown in Section 6.